

Pranay Grandhi

pranaygrandhi9@gmail.com | +91-8297380938 | github.com/pranaygrandhi |
linkedin.com/in/pranay-grandhi

Career Objective

Computer Science (Cyber Security) undergraduate with a strong foundation in network security, cryptography, vulnerability assessment, and secure system design. Passionate about solving complex engineering challenges and applying cybersecurity principles to mission-critical systems. Highly interested in space technology, aerospace engineering, rocket systems, and embedded technologies. Currently contributing to the IN-SPACe Rocketry Student Competition as a Critical Design Review (CDR) qualified team member. Seeking opportunities to combine cybersecurity, software development, and engineering skills to build secure and innovative aerospace systems.

Education

SRM Institute of Science and Technology

2024 – Present

B.Tech – Computer Science and Engineering (Cyber Security)

CGPA: **9.27**

Experience

Network Security Associate Intern (Fortinet)

Jul 2024 – Oct 2024

EduSkills

- Analyzed network traffic and identified vulnerabilities using industry-standard security tools.
- Applied firewall, IDS, and network security concepts to strengthen cybersecurity knowledge.

Ethical Hacking Intern

Jul 2025 – Oct 2025

EduSkills

- Conducted vulnerability assessments and penetration testing on simulated environments.
- Identified security weaknesses and proposed mitigation strategies following industry best practices.

Projects

IN-SPACe Rocketry Student Competition (Critical Design Review Qualifier)

- Successfully qualified for the **Critical Design Review (CDR)** after clearing the **Preliminary Design Review (PDR)** in the national-level IN-SPACe Rocketry Student Competition.
- Collaborated with a multidisciplinary engineering team in the design and development of a student-built sounding rocket.
- Contributed to subsystem design, engineering calculations, system integration, and technical documentation following aerospace engineering standards.
- Participated in design validation, risk assessment, trade-off analysis, and preparation of technical reports and presentations for the CDR evaluation.

- Strengthened knowledge of rocket systems, systems engineering, aerospace design principles, and space technologies through practical project experience.

Security Analysis of Internet-Based Voice and Video Communication Infrastructure

- Analyzed VoIP protocols (SIP and RTP) across 100+ packets using Wireshark.
- Identified Man-in-the-Middle (MITM) and replay attacks.
- Implemented Secure RTP (SRTP) and TLS encryption for secure communication.
- Performed packet-level traffic analysis and evaluated communication security.

Practical Attacks and Countermeasures for Cryptosystems

- Simulated brute-force, timing, and cryptographic attacks.
- Designed secure countermeasures using robust key management techniques.
- Improved system security by reducing vulnerability exposure.

Security Analysis of CPU–GPU Coupled Architectures

- Investigated vulnerabilities in heterogeneous CPU–GPU architectures.
- Identified memory leakage and side-channel attack risks.
- Proposed secure memory isolation techniques to improve system resilience.

Technical Skills

Programming Languages: Python, C, C++, Java

Cybersecurity: Ethical Hacking, Vulnerability Assessment, Threat Analysis, Network Security, Cryptography

Tools: Wireshark, Nmap, Burp Suite, Linux

Developer Tools: GitHub, Visual Studio Code

Core Concepts: Secure Communication, Information Security, Network Security, System Security

Certifications

- Fortinet Certified Associate (FCA)
- Fortinet Certified Fundamentals (FCF)
- Cisco Network Defence
- ISO/IEC 27001:2022 Lead Auditor